

Tema 2 - Utilizarea modelelor neurale pentru clasificarea imaginilor

1. Descrierea seturilor de date

1.1 "Vai de pielea mea" — clasificare alunate dermatoscopice

Setul contine **10 015 imagini color** de 130×100 pixeli, organizate in 7 clase de leziuni cutanate. Setul este puternic dezechilibrat — clasa **nv** (nevi melanocitari) reprezinta majoritarul absolut.

Clasa	Descriere	Train	Test
akiec	Actinic keratosis / boala Bowel	~327	65
bcc	Carcinom bazocelular	~514	103
bkl	Leziuni benigne de tip keratoza	~1099	220
df	Dermatofibrom	~115	23
mel	Melanom	~1113	222
nv	Nevi melanocitari (alunate obisnuite)	~6705	1341
vasc	Leziuni vasculare	~142	29

Normalizare (RGB): mean = [0.7646, 0.5464, 0.5712], std = [0.1410, 0.1532, 0.1706]

Split: Train = 6009, Test local = 2003, Kaggle test = 2003.

1.2 "You're on candid camera" — recunoastere expresii faciale

Setul contine **15 339 imagini color** de 100×100 pixeli, organizate in 7 clase de emotii de baza. Dezechilibru semnificativ: Happiness ~38.9%, Fear ~2.3%.

Clasa	Emotie	Train	Local Test	Remote Test
1	Surprise	1097 (10.5%)	193	329
2	Fear	239 (2.3%)	42	74
3	Disgust	610 (5.8%)	107	160
4	Happiness	4057 (38.9%)	715	1185
5	Sadness	1685 (16.1%)	297	478
6	Anger	600 (5.8%)	105	162
7	Neutral	2146 (20.6%)	378	680

Split: Train = 10 434, Local Test = 1 837, Remote Test = 3 068.

2. Vizualizarea augmentarilor (3.1)

Biblioteca folosita: **Albumentations**, integrata printr-un wrapper ce aplica augmentarile inainte de normalizarea torchvision. Augmentarile sunt aplicate **exclusiv pe setul de antrenare**.

2.1 Augmentari — "Vai de pielea mea"

Mole-urile nu au o orientare naturala — rotatiile la orice unghi sunt valide. ColorJitter este intentionat subtil pentru a nu sterge culorile relevante diagnosticului (melanom vs. nev).

Tip augmentare	Transformari aplicate
Geometrice	HorizontalFlip(p=0.5), VerticalFlip(p=0.5), RandomRotate90(p=0.5), Affine(translate $\pm 5\%$, scale 0.9–1.1, rotate $\pm 30^\circ$, p=0.5)
Culori	ColorJitter(brightness=0.15, contrast=0.15, saturation=0.1, hue=0.03, p=0.7), GaussianBlur(3–5px, p=0.2), CLAHE(p=0.2)
Toate	Geometrice + Culori

Augmentari — 'Vai de pielea mea'

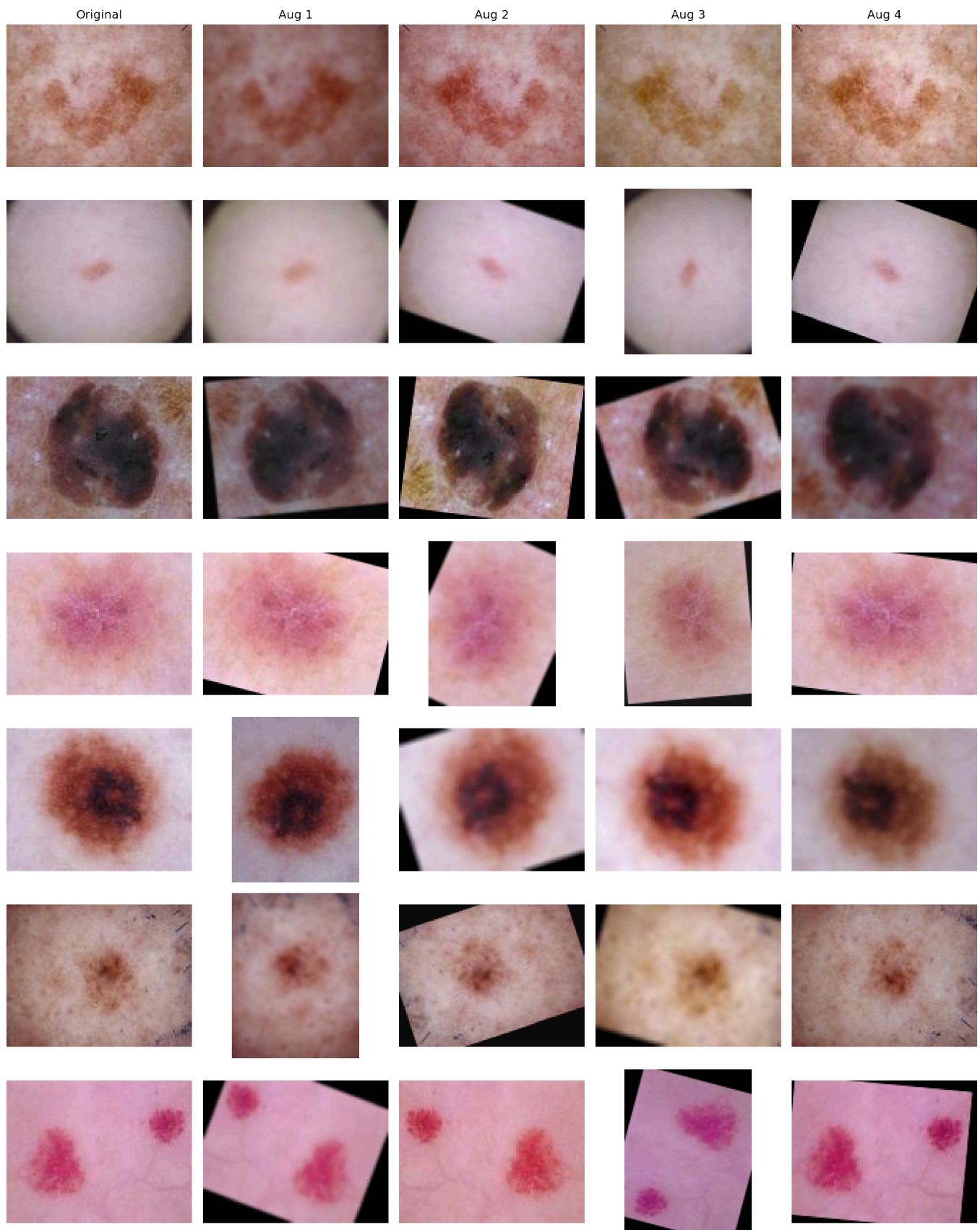


Fig. 1 — Exemple augmentate per clasa, dataset "Vai de pielea mea"

Observatie: Augmentarile geometrice agresive (flip vertical, rotatie 90°) sunt justificate deoarece leziunile dermatoscopice sunt fotografiate din unghiuri arbitrare. ColorJitter minimal pastreaza structurile pigmentare (culori specifice melanomului).

2.2 Augmentari — "You're on candid camera"

Nu se aplica VerticalFlip — fetele nu apar cu capul in jos. Rotatiile sunt limitate la $\pm 15^\circ$ pentru a pastra alinierea faciala. ColorJitter mai agresiv deoarece variabilitatea de iluminare in conditii reale este ridicata.

Tip augmentare	Transformari aplicate
Geometrice	HorizontalFlip($p=0.5$), Affine(translate $\pm 5\%$, scale 0.9–1.1, rotate $\pm 15^\circ$, $p=0.5$), RandomCrop(88×88 , $p=0.5$) + PadIfNeeded(96×96)
Culori	ColorJitter(brightness=0.3, contrast=0.3, saturation=0.2, hue=0.05, $p=0.8$), GaussianBlur(3–5px, $p=0.3$), ToGray($p=0.1$)
Toate	Geometrice + Culori

Augmentari — 'You're on candid camera'



Fig. 2 — Exemple augmentate per clasa, dataset "You're on candid camera"

Observatie: ToGray($p=0.1$) simuleaza conditii de iluminare monocromatica, frecvente in imagini din surse diverse. Augmentarile de culoare mai intense sunt justificate de variabilitatea mare de iluminare si

echipamente.

3. Arhitectura MLP (3.2)

3.1 Descrierea arhitecturii

Imaginile sunt redimensionate la **64×64×3** → aplatizate la un vector de **12 288** elemente. Arhitectura are 3 straturi ascunse cu dimensiuni descrescatoare, BatchNorm1d dupa fiecare strat liniar si Dropout cu rate descrescatoare.

```
Flatten → Linear(12288, 1024) → BN1d → ReLU → Dropout(0.4)
      → Linear(1024, 512) → BN1d → ReLU → Dropout(0.3)
      → Linear(512, 256) → BN1d → ReLU → Dropout(0.2)
      → Linear(256, 7)
```

Parametru	Valoare
Parametri totali	13 245 447
Epoci	20
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)
Scheduler	CosineAnnealingLR (T_max=20)
Batch size	64
Input size	64×64 px

Motivatie: BatchNorm1d accelereaza convergenta si reduce sensibilitatea la initializare. Dropout descrescator (0.4 → 0.2) aplica regularizare mai agresiva la primele straturi (mai multi parametri, risc mai mare de overfitting) si mai blanda aproape de iesire.

3.2 Rezultate MLP — "Vai de pielea mea"

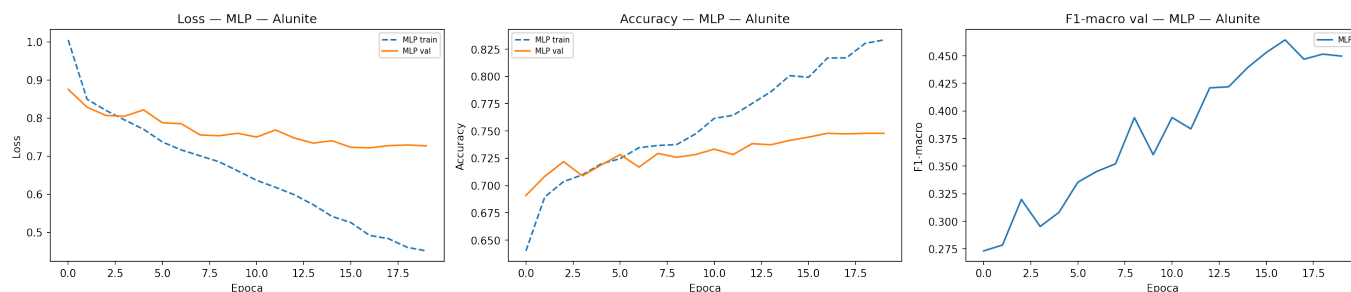


Fig. 3 — Curbe loss, accuracy si F1-macro val — MLP, Alunite

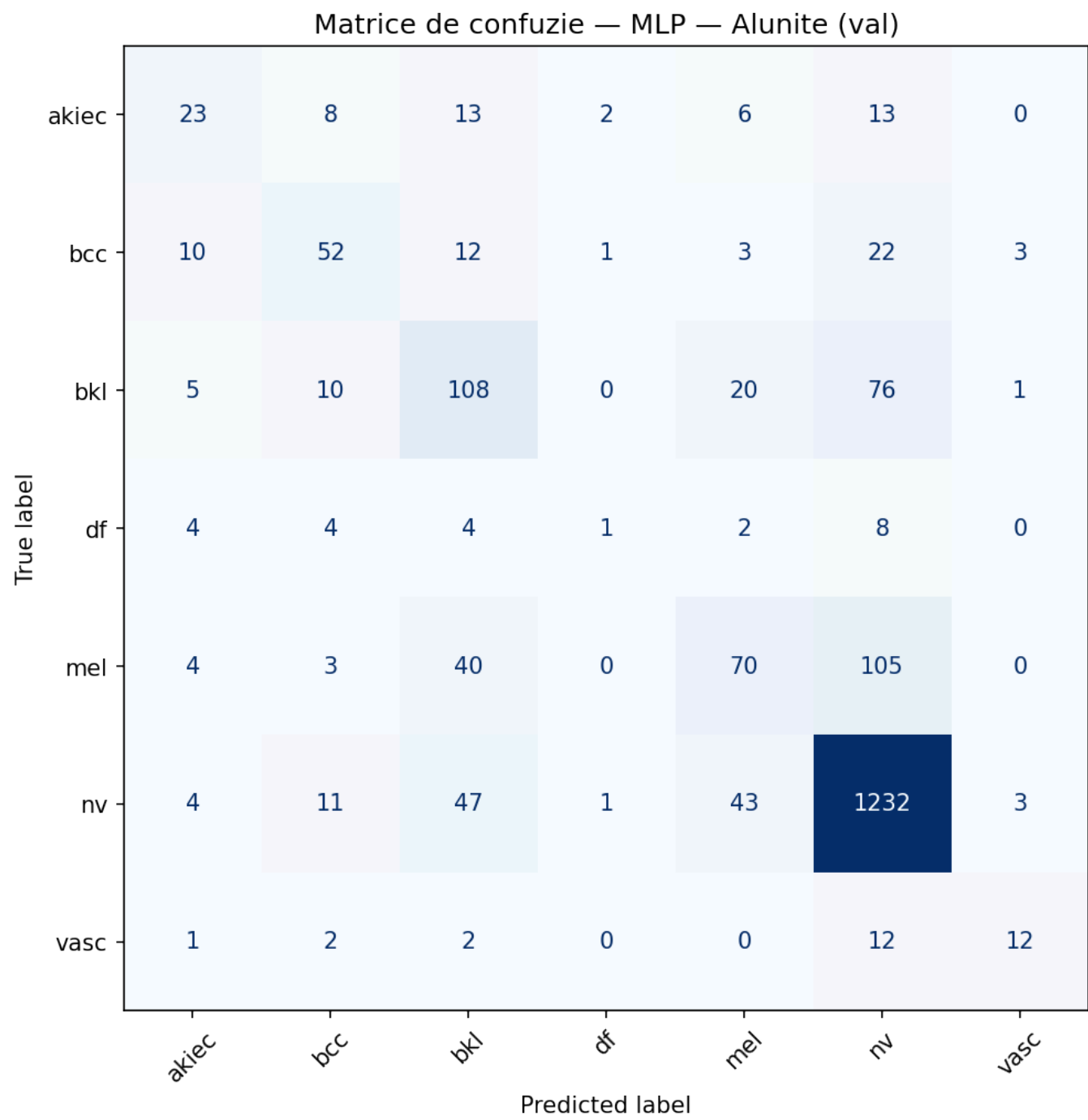


Fig. 4 — Matrice de confuzie — MLP, Alunite

Metrica	Valoare		
Accuracy	0.7479		
F1-macro	0.4644		
F1-micro	0.7479		
Clasa	Precision	Recall	F1
akiec	0.45	0.35	0.40
bcc	0.58	0.50	0.54
bkl	0.48	0.49	0.48

Clasa	Precision	Recall	F1
df	0.20	0.04	0.07
mel	0.49	0.32	0.38
nv	0.84	0.92	0.88
vasc	0.63	0.41	0.50

Observatie: Accuracy ridicata (0.75) este influentata masiv de clasa majoritara **nv**. F1-macro de 0.46 reflecta performanta slaba pe clasele minoritare (df: F1=0.07, mel: F1=0.38). MLP-ul sufera de dezechilibrul de clase, predictand predominant **nv**.

3.3 Rezultate MLP — "You're on candid camera"

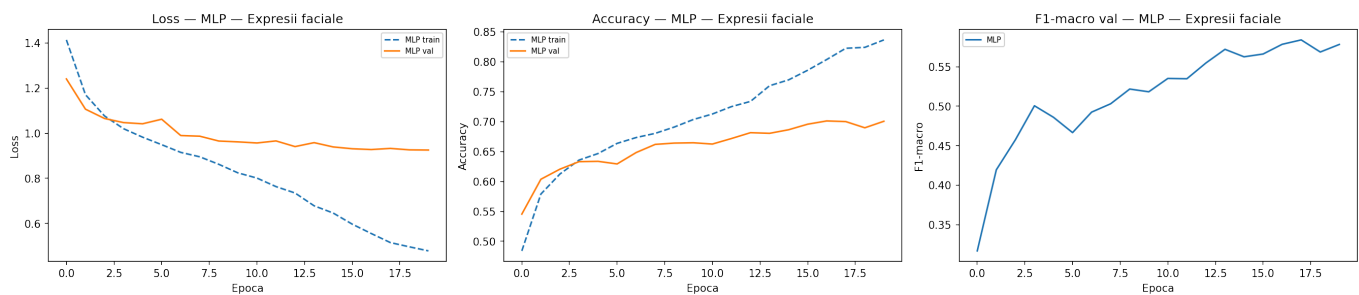


Fig. 5 — Curbe loss, accuracy si F1-macro val — MLP, Expresii faciale

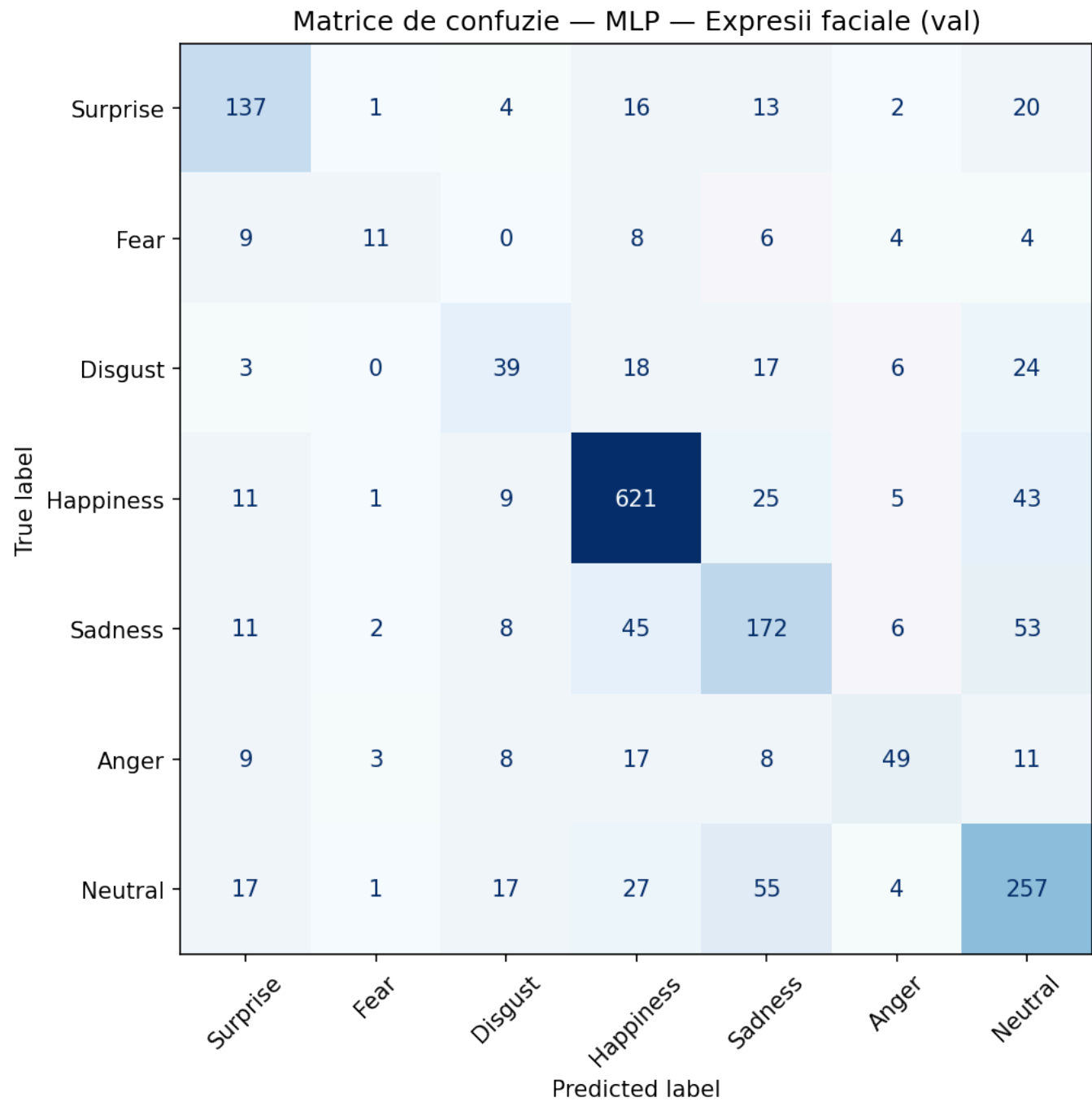


Fig. 6 — Matrice de confuzie — MLP, Expresii faciale

Metrica	Valoare		
Accuracy	0.7001		
F1-macro	0.5840		
F1-micro	0.7001		
Clasa	Precision	Recall	F1
Surprise	0.70	0.71	0.70
Fear	0.58	0.26	0.36
Disgust	0.46	0.36	0.41

Clasa	Precision	Recall	F1
Happiness	0.83	0.87	0.85
Sadness	0.58	0.58	0.58
Anger	0.64	0.47	0.54
Neutral	0.62	0.68	0.65

Observatie: Performanta mai echilibrata pe expresii faciale (F1-macro=0.584) comparativ cu alunite. **Fear** ramane dificila (F1=0.36) din cauza reprezentarii reduse in antrenare (~2.3%). **Happiness** beneficiaza clar de suportul majoritar (F1=0.85).

4. Arhitectura CNN (4.3)

4.1 Descrierea arhitecturii

Arhitectura CNN este **comuna** ambelor seturi de date, cu singurul parametru variabil **first_kernel** (3×3 pentru fete, 5×5 pentru alunite). Partea convolutionala consta in **4 blocuri Conv → BN → ReLU → Pool**, finalizata cu **AdaptiveAvgPool2d(1)** — agnostic la rezolutia de intrare. Clasificatorul are **2 straturi FC**.

```
Conv(3, 32, first_kernel) → BN → ReLU → MaxPool(2)    [Block 1 – first_kernel configurabil]
Conv(32, 64, 3) → BN → ReLU → MaxPool(2)    [Block 2]
Conv(64, 128, 3) → BN → ReLU → MaxPool(2)    [Block 3]
Conv(128, 256, 3) → BN → ReLU → AdaptiveAvgPool(1) [Block 4 – GAP]
Flatten → Dropout(0.5) → Linear(256, 128) → ReLU → Dropout(0.3) →
Linear(128, 7)
```

Parametru	Valoare
Parametri totali	422 695
Epoci	20
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)
Scheduler	CosineAnnealingLR (T_max=20)
Batch size	64 (drop_last=True)
Input alunite	96×96 px, first_kernel=5
Input fete	96×96 px, first_kernel=3

Motivatie: **first_kernel=5** pentru alunite ofera un receptive field mai mare initial, potrivit structurilor pigmentare de scara mai mare (asimetrie, margini neregulate). **first_kernel=3** pentru fete capteaza trasaturi fine (marginii ochi, gura, sprancene). **AdaptiveAvgPool2d(1)** inlocuieste stratul Flatten clasic, facand arhitectura agnostica la rezolutie.

4.2 Rezultate CNN — "Vai de pielea mea"

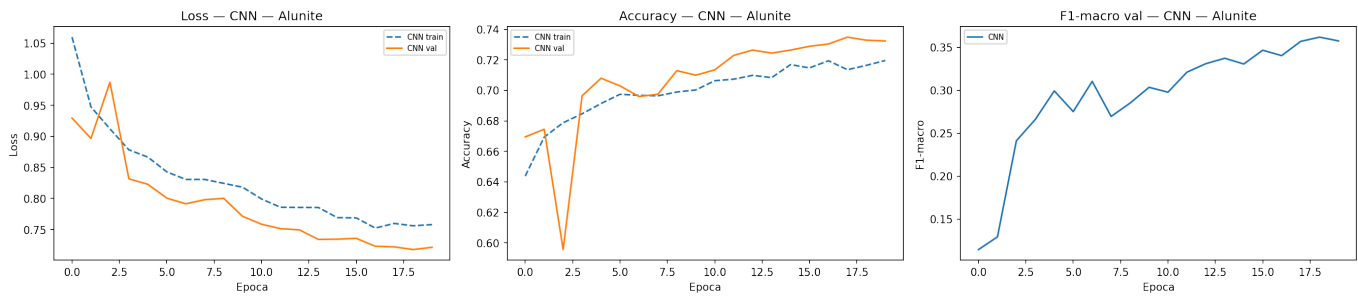


Fig. 7 — Curbe loss, accuracy si F1-macro val — CNN, Alunite

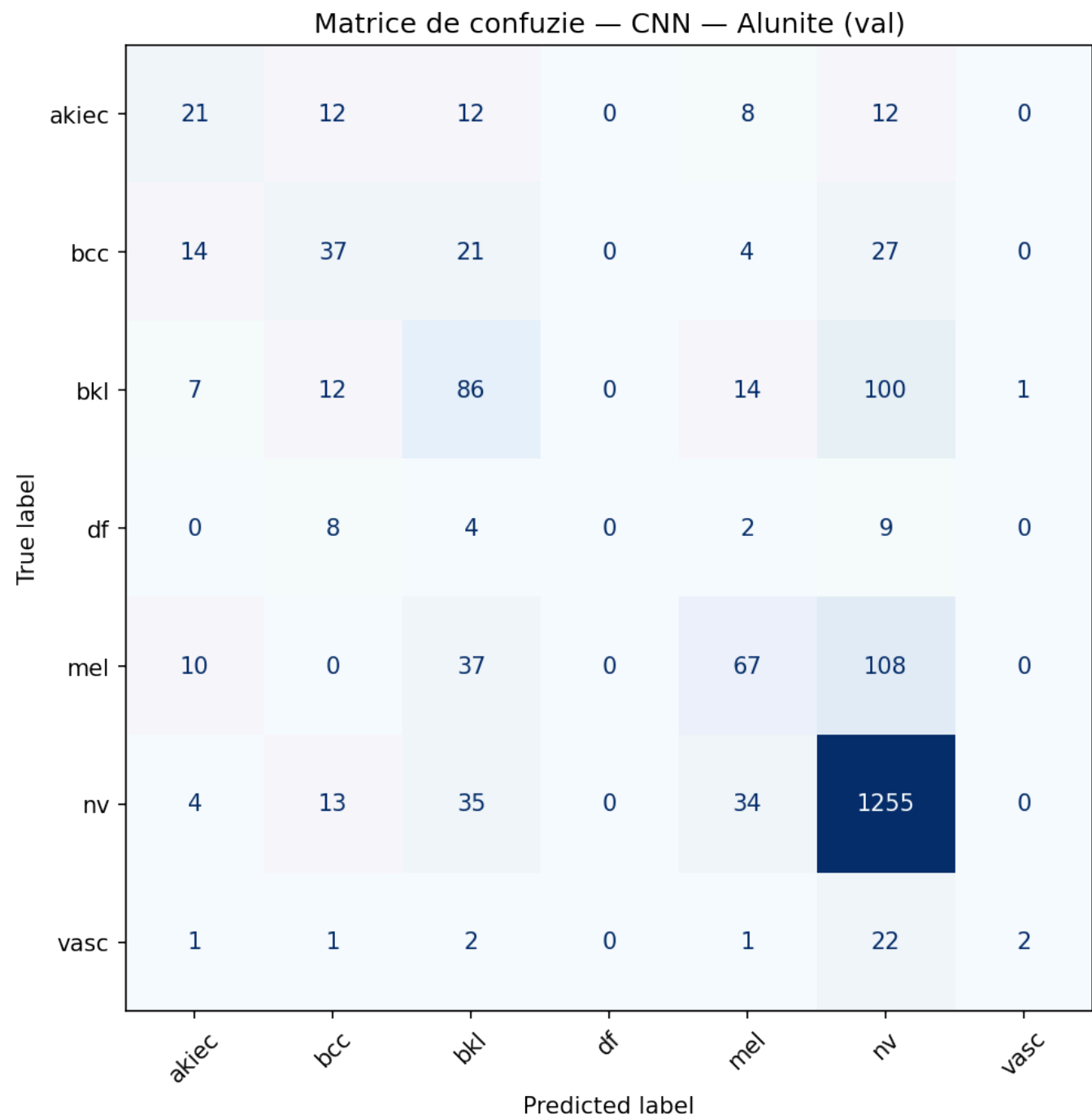


Fig. 8 — Matrice de confuzie — CNN, Alunite

Metrica	Valoare		
Accuracy	0.7329		
F1-macro	0.3619		
F1-micro	0.7329		
Clasa	Precision	Recall	F1
akiec	0.37	0.32	0.34
bcc	0.45	0.36	0.40
bkl	0.44	0.39	0.41
df	0.00	0.00	0.00
mel	0.52	0.30	0.38
nv	0.82	0.94	0.87
vasc	0.67	0.07	0.12

Observatie: CNN-ul antrenat de la zero pe alunite sufera mai mult decat MLP-ul pe clasele minoritare (df: F1=0.00, vasc: F1=0.12). Dezechilibrul de clase este mai evident — modelul converge spre predictia **nv**. F1-macro de 0.36 indica underfit pe clasele rare fara tehnici de reechilibrare.

4.3 Rezultate CNN — "You're on candid camera"

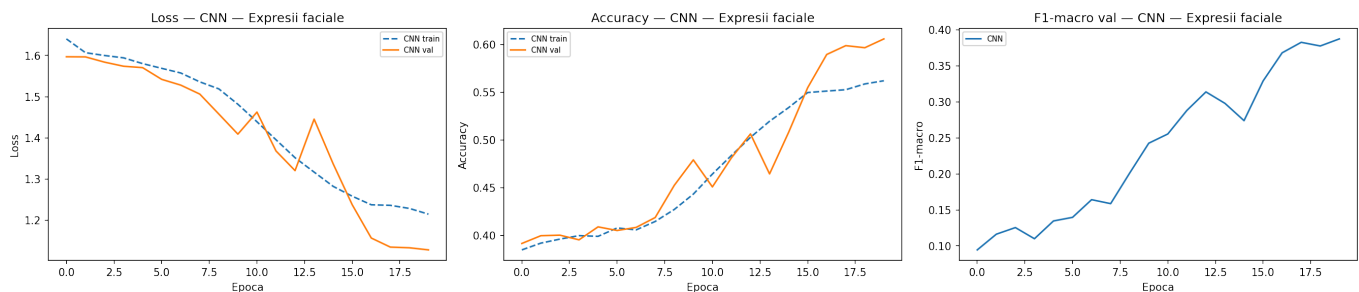


Fig. 9 — Curbe loss, accuracy si F1-macro val — CNN, Expresii faciale

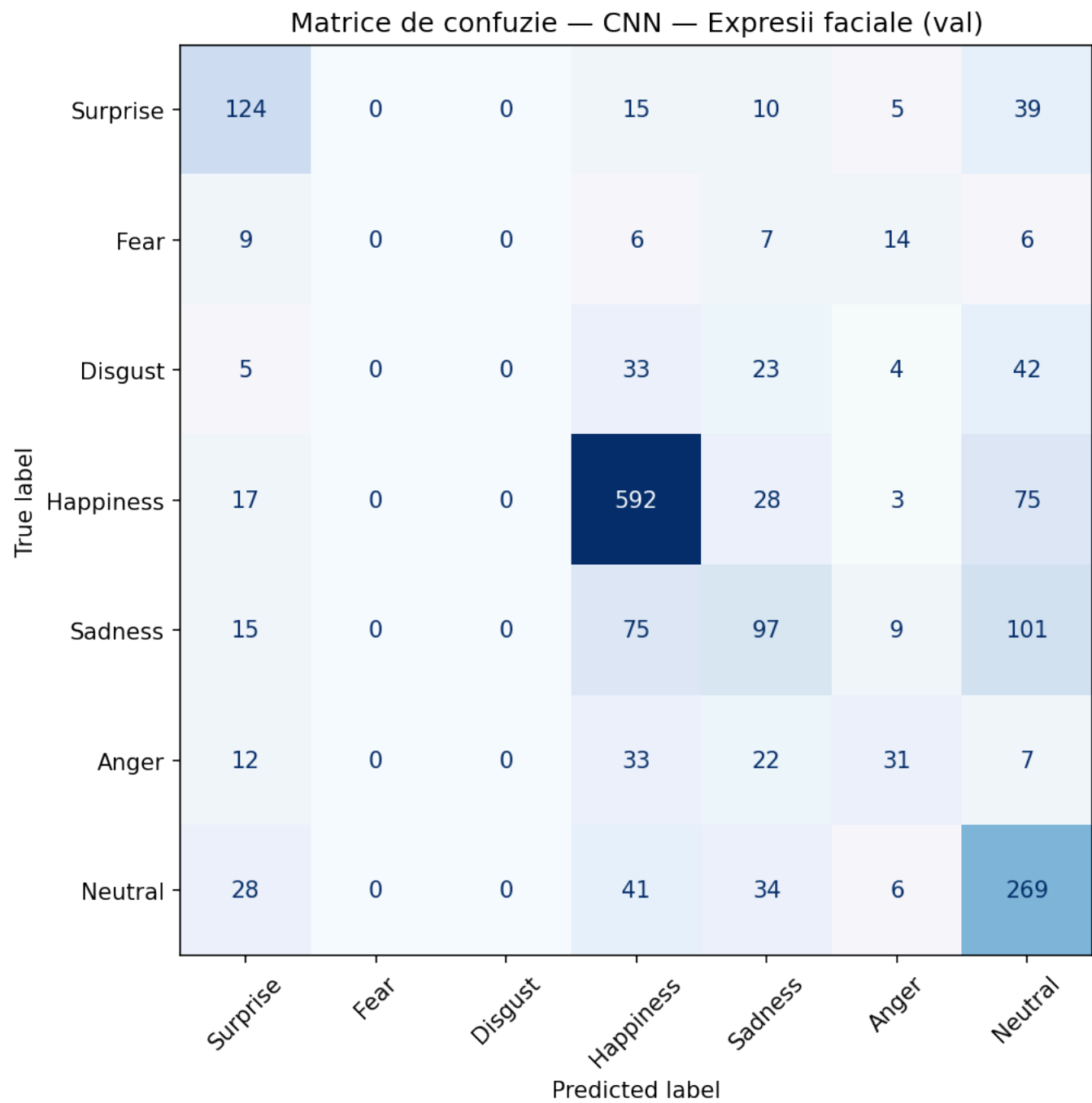


Fig. 10 — Matrice de confuzie — CNN, Expresii faciale

Metrica	Valoare
Accuracy	0.6059
F1-macro	0.3873
F1-micro	0.6059

Clasa	Precision	Recall	F1
Surprise	0.59	0.64	0.62
Fear	0.00	0.00	0.00
Disgust	0.00	0.00	0.00

Clasa	Precision	Recall	F1
Happiness	0.74	0.83	0.78
Sadness	0.44	0.33	0.37
Anger	0.43	0.30	0.35
Neutral	0.50	0.71	0.59

Observatie: CNN-ul antrenat de la zero pe expresii faciale nu reuseste sa invete clasele **Fear** si **Disgust** (F1=0.00 ambele), indicand un caz sever de overfit pe clasele majoritare. Comparativ cu MLP-ul, CNN-ul performa mai slab fara augmentare si tehnici de reechilibrare — confirma nevoia de transfer learning.

5. Fine-tuning ResNet-18 (3.4)

5.1 Procedura de fine-tuning

S-a folosit **ResNet-18 pre-antrenat pe ImageNet**. Strategia de dezghetare: **layer1** si **layer2** ramanana inghetate; **layer3**, **layer4** si capul **fc** sunt antrenate. Capul original (Linear(512, 1000)) a fost inlocuit cu **Dropout(0.5) → Linear(512, 7)**.

Parametru	Valoare
Arhitectura	ResNet-18 (ImageNet pretrained)
Straturi inghetate	layer1, layer2
Straturi antrenate	layer3, layer4, fc
Epoci	40 (3 warmup + 37 cosine decay)
Optimizer	Adam (lr=1e-3, weight_decay=1e-4)
Scheduler	LR warmup liniar (3 ep) + CosineAnnealingLR
Batch size	32
Input	224×224 px (redimensionat din 96/100 px)
Normalizare	Statistici ImageNet [0.485, 0.456, 0.406] / [0.229, 0.224, 0.225]

5.2 Rezultate Fine-tuning — "Vai de pielea mea"

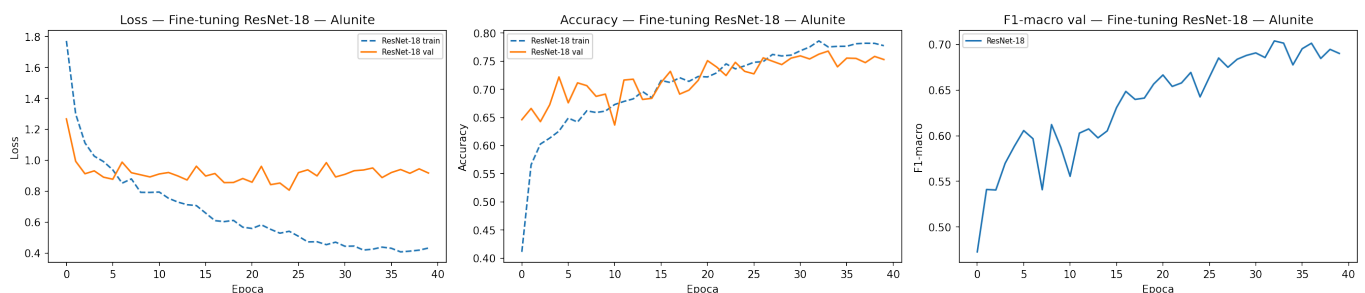


Fig. 11 — Curbe loss, accuracy si F1-macro val — ResNet-18, Alunite

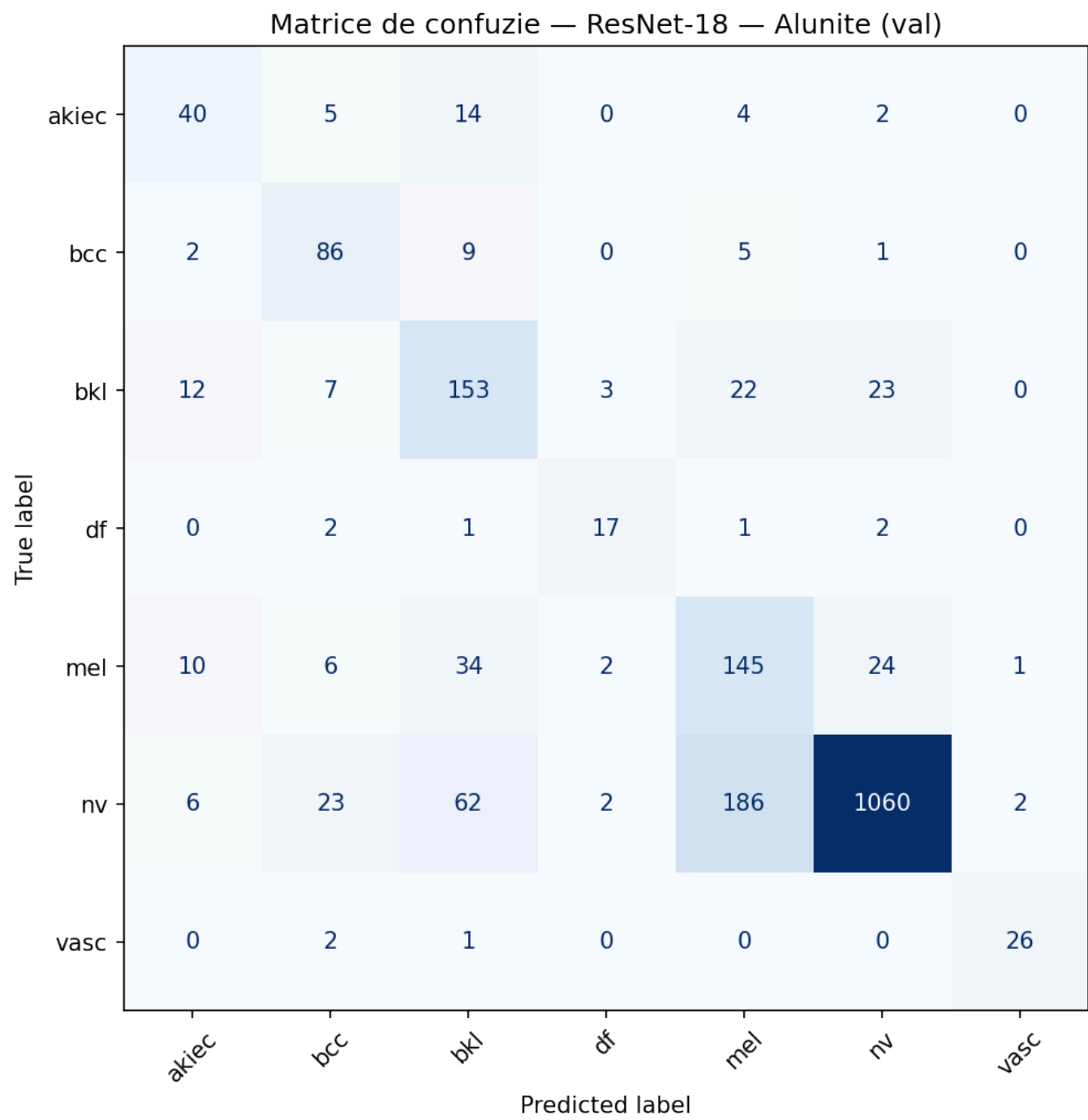


Fig. 12 — Matrice de confuzie — ResNet-18, Alunite

Metrica	Valoare		
Accuracy	0.7624		
F1-macro	0.7039		
F1-micro	0.7624		
Clasa	Precision	Recall	F1
akiec	0.57	0.62	0.59
bcc	0.66	0.83	0.74
bkl	0.56	0.70	0.62

Clasa	Precision	Recall	F1
df	0.71	0.74	0.72
mel	0.40	0.65	0.50
nv	0.95	0.79	0.86
vasc	0.90	0.90	0.90

Observatie: Salt major fata de CNN antrenat de la zero — F1-macro creste de la 0.36 la 0.70. Clasele minoritare **df** (F1=0.72) si **vasc** (F1=0.90) sunt acum bine recunoscute, demonstrand ca feature-urile ImageNet sunt transferabile si pe imagini medicale. Singurul punct slab ramas este **mel** (F1=0.50) — melanomul este vizual similar cu nevii.

5.3 Rezultate Fine-tuning — "You're on candid camera"

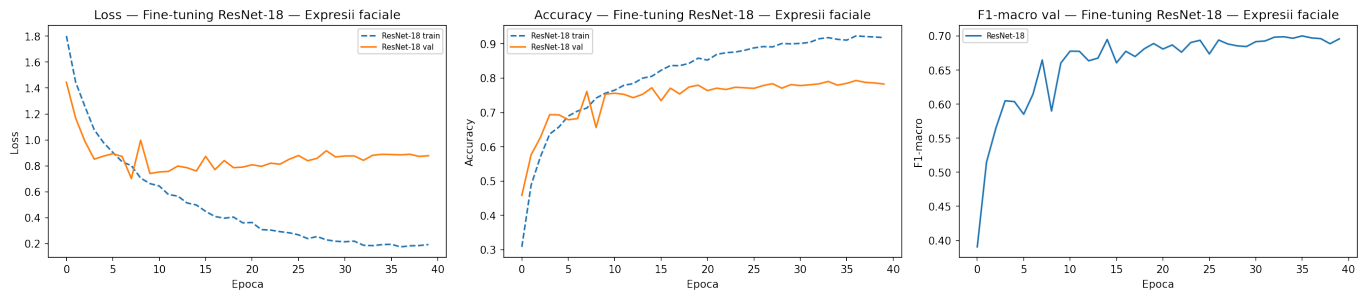


Fig. 13 — Curbe loss, accuracy si F1-macro val — ResNet-18, Expresii faciale

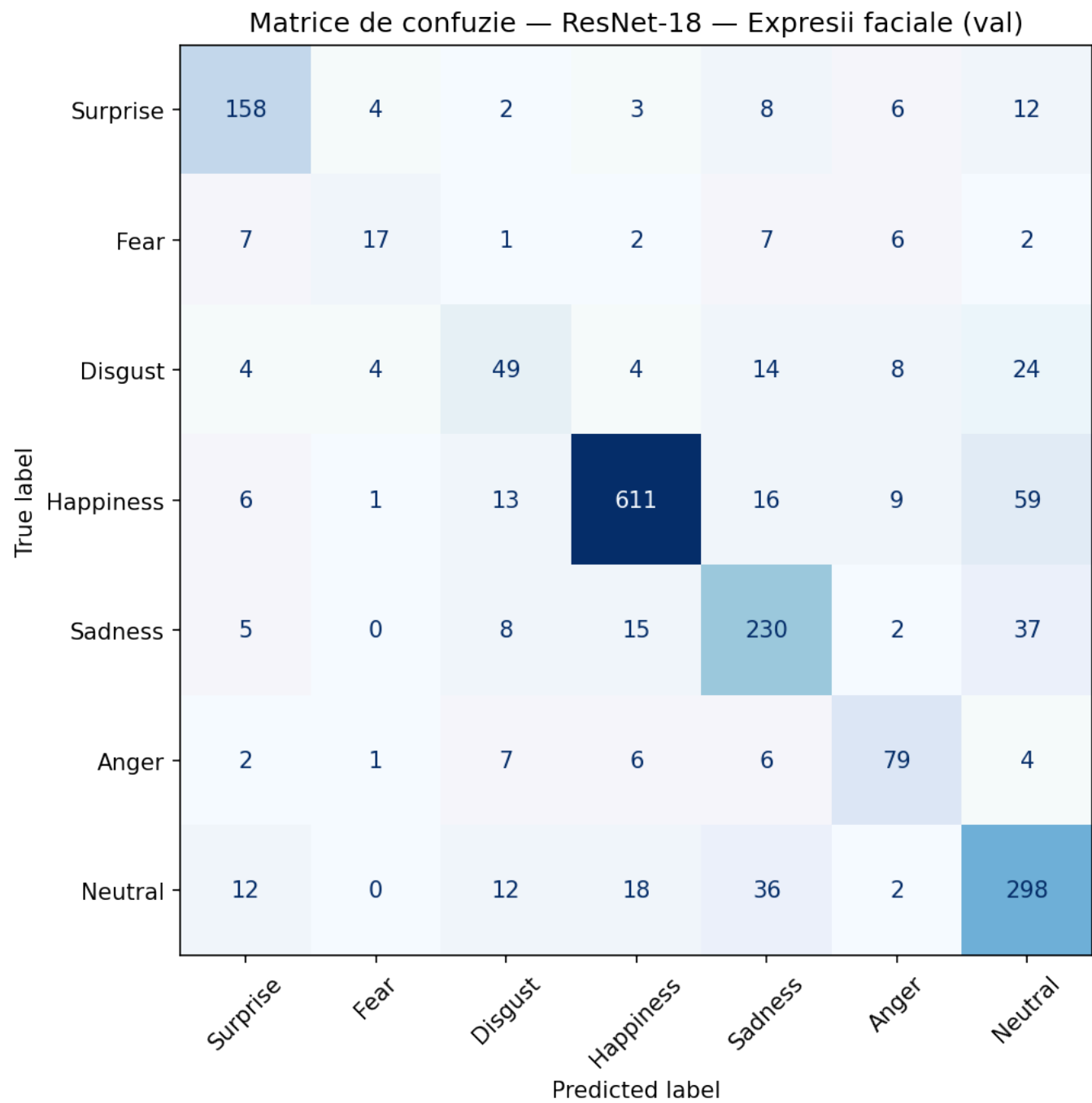


Fig. 14 — Matrice de confuzie — ResNet-18, Expresii faciale

Metrica	Valoare
Accuracy	0.7850
F1-macro	0.7001
F1-micro	0.7850

Clasa	Precision	Recall	F1
Surprise	0.81	0.82	0.82
Fear	0.63	0.40	0.49
Disgust	0.53	0.46	0.49

Clasa	Precision	Recall	F1
Happiness	0.93	0.85	0.89
Sadness	0.73	0.77	0.75
Anger	0.71	0.75	0.73
Neutral	0.68	0.79	0.73

Observatie: Imbunatatire dramatica fata de CNN de la zero (F1-macro 0.39 → 0.70). Toate clasele au acum $F1 > 0$, inclusiv **Fear** si **Disgust** care anterior erau 0.00. **Fear** ramane cea mai grea clasa ($F1=0.49$) din cauza reprezentarii reduse.

6. Rezultate prin ablatii (3.5)

6.1 Ablatie MLP — cu / fara Dropout

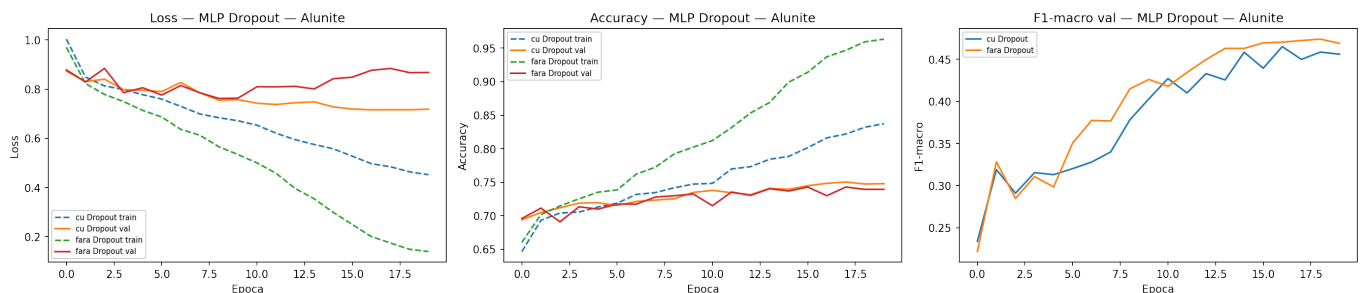


Fig. 15 — Ablatie Dropout — MLP, Alunite

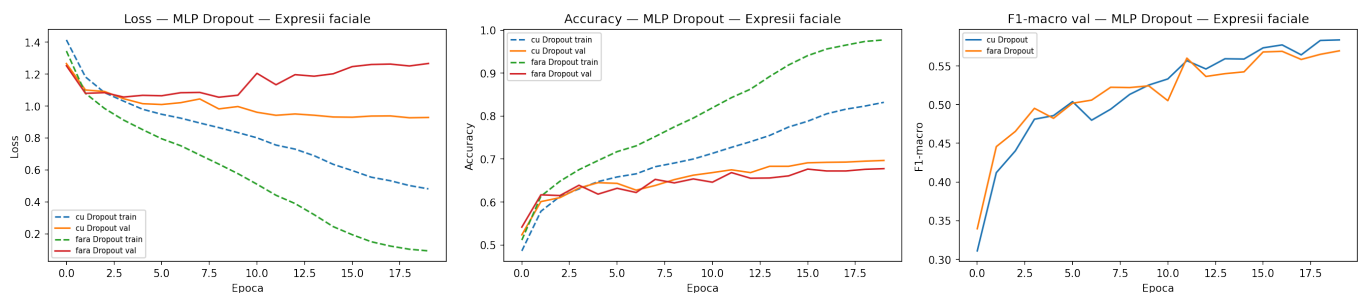


Fig. 16 — Ablatie Dropout — MLP, Expressii faciale

Experiment	Dataset	Best val acc	Best F1-macro
MLP cu Dropout	Alunite	0.7504	0.4653
MLP fara Dropout	Alunite	0.7429	0.4741
MLP cu Dropout	Fete	0.6968	0.5836
MLP fara Dropout	Fete	0.6777	0.5695

Observatie: Dropout imbunatateste consistent accuracy-ul si F1-macro pe ambele seturi de date. Diferenta este mai pronuntata pe expresii faciale (+0.02 acc, +0.014 F1) decat pe alunite. Fara Dropout, modelul face mai mult overfit pe exemplele majoritare — curbele de val pierd mai repede din stabilitate.

6.2 Ablatie CNN — cu / fara BatchNorm

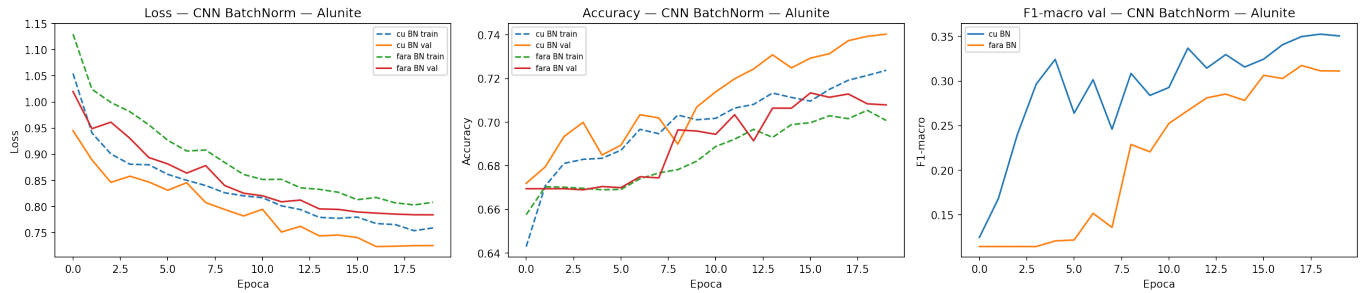


Fig. 17 — Ablatie BatchNorm — CNN, Alunite

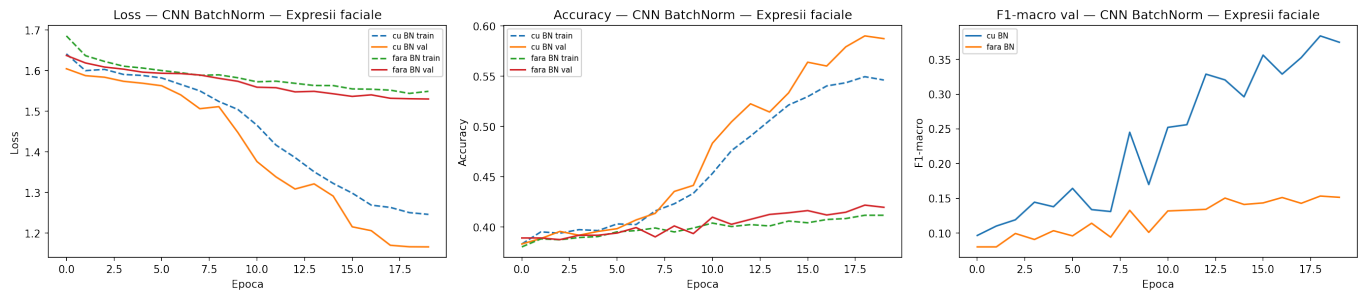


Fig. 18 — Ablatie BatchNorm — CNN, Expresii faciale

Experiment	Dataset	Best val acc	Best F1-macro
CNN cu BN	Alunite	0.7404	0.3526
CNN fara BN	Alunite	0.7134	0.3175
CNN cu BN	Fete	0.5901	0.3839
CNN fara BN	Fete	0.4219	0.1533

Observatie: Impactul BatchNorm este extrem de vizibil pe expresii faciale — fara BN, F1-macro colapseaza la 0.15 (de la 0.38 cu BN). Pe alunite diferenta este mai mica, dar consistenta. Fara BN, antrenarea devine instabila si convergenta mai lenta, confirma rolul esential al normalizarii in CNN-uri mici antrenate pe date putine.

6.3 Ablatie CNN — configuratii de augmentare

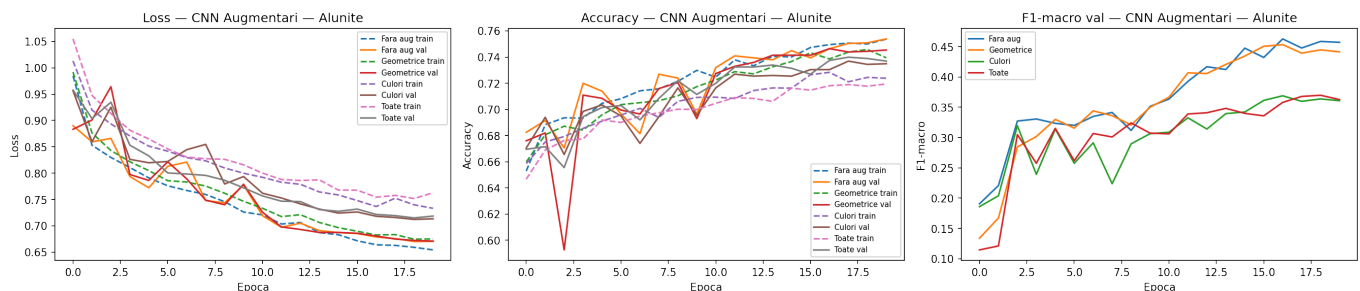


Fig. 19 — Ablatie augmentari — CNN, Alunite

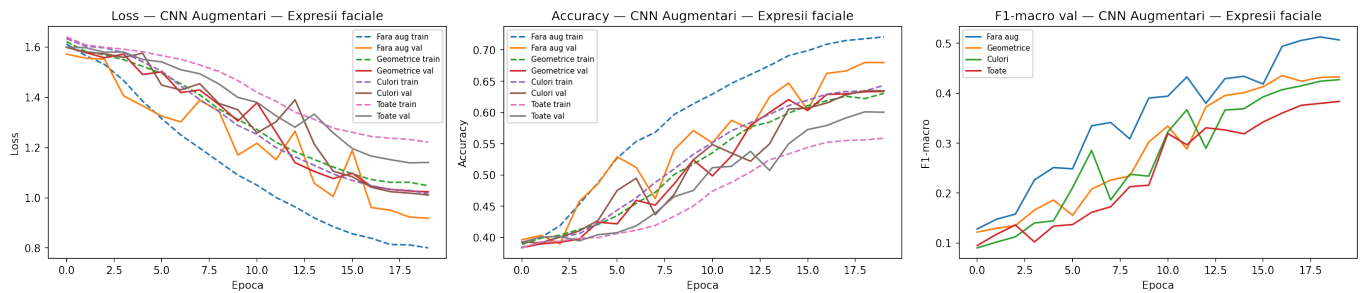


Fig. 20 — Ablatie augmentari — CNN, Expresii faciale

Experiment	Dataset	Best val acc	Best F1-macro
CNN fara aug	Alunite	0.7539	0.4629
CNN aug geometric	Alunite	0.7464	0.4534
CNN aug culori	Alunite	0.7369	0.3690
CNN aug toate	Alunite	0.7399	0.3699
CNN fara aug	Fete	0.6799	0.5126
CNN aug geometric	Fete	0.6336	0.4352
CNN aug culori	Fete	0.6347	0.4270
CNN aug toate	Fete	0.6010	0.3835

Observatie: Contrar asteptarilor, augmentarea nu imbunatateste CNN-ul antrenat de la zero pe aceste seturi. Pe alunite, configuratia fara augmentare obtine cel mai mare F1-macro (0.46), iar pe fete diferenta este si mai mare (0.51 vs 0.38). Explicatia probabila: CNN-ul cu ~400K parametri si 20 epoci nu are suficienta capacitate si timp de antrenare pentru a beneficia de variabilitatea introdusa de augmentari — augmentarea face exemplele mai greu de invatat, dar modelul nu este destul de puternic sa generalizeze din aceasta variabilitate. Augmentarea isi arata beneficiul clar in contextul fine-tuning-ului (modele mai puternice, mai multe epoci).

6.4 Ablatie — Tehnici de tratare a dezechilibrului de clase

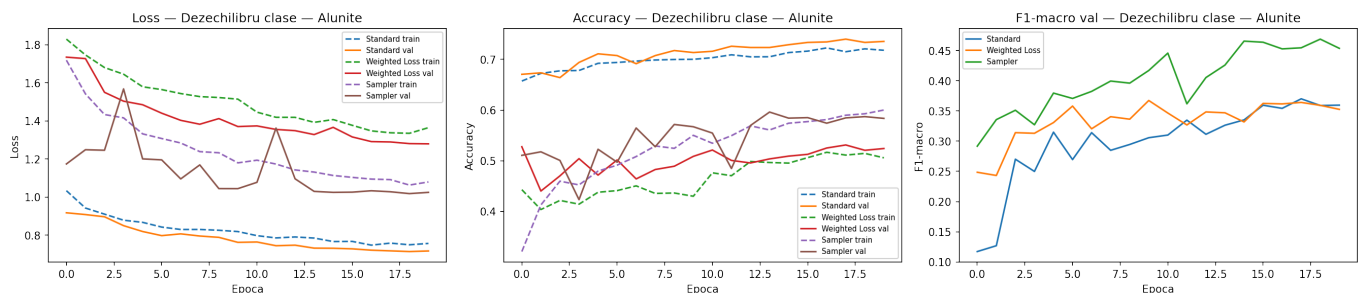


Fig. 21 — Ablatie dezechilibru clase — CNN, Alunite

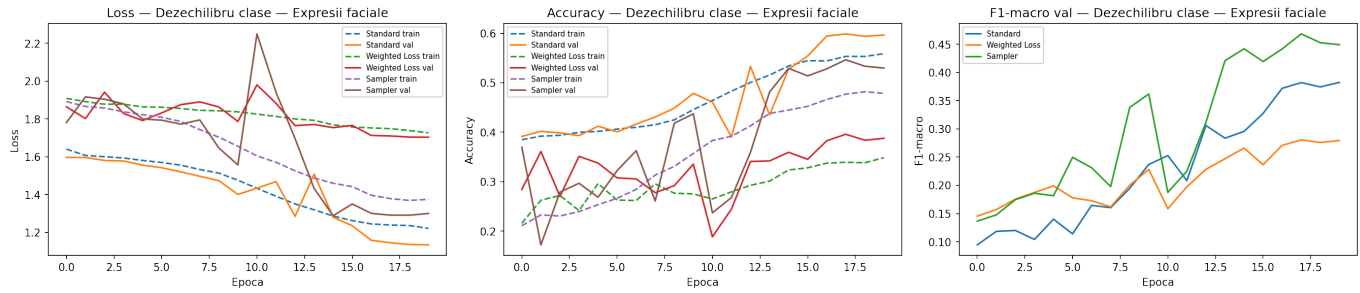


Fig. 22 — Ablatie dezechilibru clase — CNN, Expresii faciale

Experiment	Dataset	Best val acc	Best F1-macro
CNN standard	Alunite	0.7399	0.3702
CNN Weighted Loss	Alunite	0.5312	0.3674
CNN Sampler	Alunite	0.5961	0.4691
CNN standard	Fete	0.5988	0.3822
CNN Weighted Loss	Fete	0.3958	0.2804
CNN Sampler	Fete	0.5465	0.4683

Observatie: WeightedRandomSampler creste F1-macro pe ambele seturi (+0.10 Alunite, +0.086 Fete) in raport cu configuratia standard, dar cu cost in accuracy globala (modelul prezice mai des clasele minoritare). Weighted Loss scade puternic accuracy fara beneficiu clar pe F1-macro, indicand ca penalizarea prin loss singura este mai greu de echilibrat fara tuning aditional al ponderilor. Sampler-ul este cea mai eficienta tehnica de reechilibrare in acest context.

6.5 Ablatie — Fine-tuning cu / fara LR Warmup

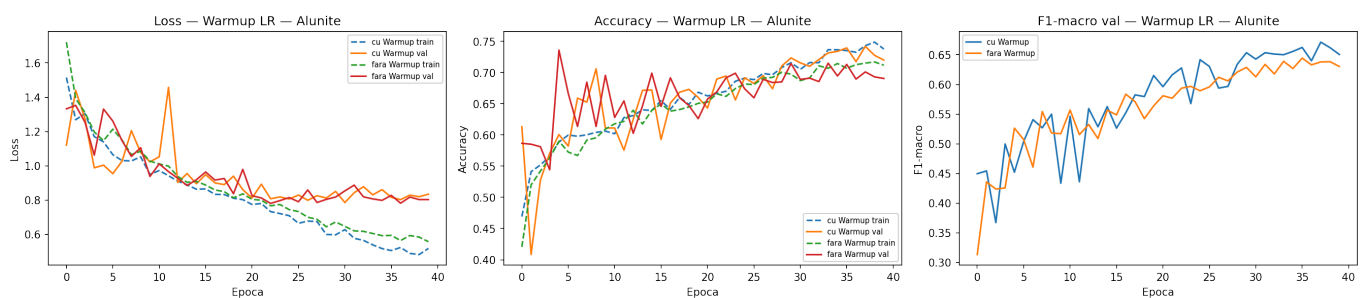


Fig. 23 — Ablatie LR Warmup — ResNet-18, Alunite

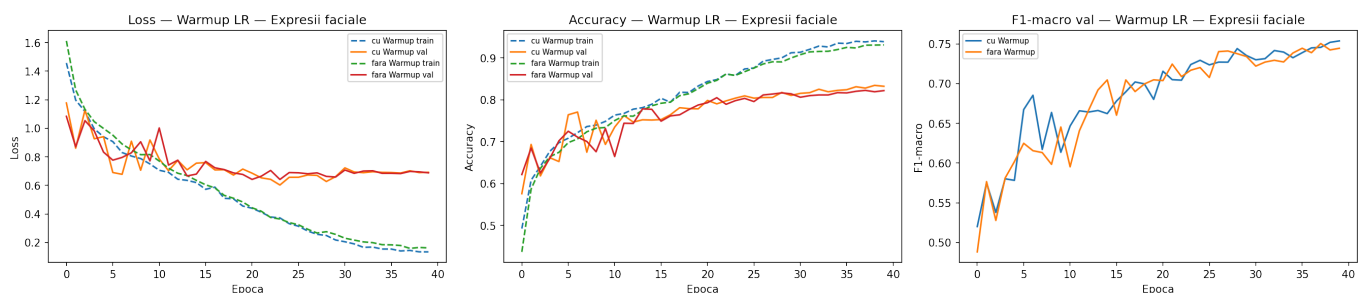


Fig. 24 — Ablatie LR Warmup — ResNet-18, Expresii faciale

Experiment	Dataset	Best val acc	Best F1-macro
FT cu Warmup	Alunite	0.7419	0.6713
FT fara Warmup	Alunite	0.7359	0.6443
FT cu Warmup	Fete	0.8345	0.7538
FT fara Warmup	Fete	0.8225	0.7505

Observatie: Warmup-ul de 3 epoci (crestere liniara LR de la 0 la lr_max) aduce imbunatatiri consistente pe ambele seturi. Pe expresii faciale, beneficiul este mai vizibil (+0.012 acc, +0.003 F1). Fara warmup, gradientii mari de la capul nou-initializat perturba ponderile pre-antrenate in primele epoci, ducand la o convergenta mai lenta si un maxim mai scazut.

6.6 Tabel comparativ — toate ablatiile

Experiment	Dataset	Best val acc	Best F1-macro
MLP cu Dropout	Alunite	0.7504	0.4653
MLP fara Dropout	Alunite	0.7429	0.4741
MLP cu Dropout	Fete	0.6968	0.5836
MLP fara Dropout	Fete	0.6777	0.5695
CNN cu BN	Alunite	0.7404	0.3526
CNN fara BN	Alunite	0.7134	0.3175
CNN cu BN	Fete	0.5901	0.3839
CNN fara BN	Fete	0.4219	0.1533
CNN fara aug	Alunite	0.7539	0.4629
CNN aug geometric	Alunite	0.7464	0.4534
CNN aug culori	Alunite	0.7369	0.3690
CNN aug toate	Alunite	0.7399	0.3699
CNN fara aug	Fete	0.6799	0.5126
CNN aug geometric	Fete	0.6336	0.4352
CNN aug culori	Fete	0.6347	0.4270
CNN aug toate	Fete	0.6010	0.3835
CNN standard	Alunite	0.7399	0.3702
CNN Weighted Loss	Alunite	0.5312	0.3674
CNN Sampler	Alunite	0.5961	0.4691

Experiment	Dataset	Best val acc	Best F1-macro
CNN standard	Fete	0.5988	0.3822
CNN Weighted Loss	Fete	0.3958	0.2804
CNN Sampler	Fete	0.5465	0.4683
FT cu Warmup	Alunite	0.7419	0.6713
FT fara Warmup	Alunite	0.7359	0.6443
FT cu Warmup	Fete	0.8345	0.7538
FT fara Warmup	Fete	0.8225	0.7505

7. Concluzii

Clasificare alunite dermatoscopice

Model	Accuracy	F1-macro	F1-micro
MLP	0.7479	0.4644	0.7479
CNN	0.7329	0.3619	0.7329
ResNet-18 (FT)	0.7624	0.7039	0.7624

- **Cel mai bun model: ResNet-18 fine-tuned** (F1-macro=0.704). Salt de aproape 2× fata de CNN de la zero pe F1-macro, demonstrand ca feature-urile ImageNet sunt direct transferabile pe imagini medicale.
- MLP depaseste CNN-ul de la zero pe F1-macro (0.46 vs 0.36), sugerand ca CNN-ul cu 20 epoci si fara augmentare/reechilibrare nu converge eficient pe clasele rare.
- Dezechilibrul de clase ramane principala provocare — clasele **df** si **vasc** cu sub 200 exemple sunt greu de invatat fara transfer learning.

Clasificare expresii faciale

Model	Accuracy	F1-macro	F1-micro
MLP	0.7001	0.5840	0.7001
CNN	0.6059	0.3873	0.6059
ResNet-18 (FT)	0.7850	0.7001	0.7850

- **Cel mai bun model: ResNet-18 fine-tuned** (F1-macro=0.700, Accuracy=0.785). Imbunatatire majora fata de MLP si CNN de la zero.
- CNN-ul de la zero esueaza complet pe **Fear** si **Disgust** (F1=0.00), evidentiind limitele antrenarii de la zero cu date putine si dezechilibrate.
- LR warmup aduce un beneficiu clar in fine-tuning — convergenta mai stabila in primele epoci protejeaza feature-urile pre-antrenate.

8. Bonus — Curriculum Learning (Self-Paced) (4.1)

8.1 Metodologie

S-a implementat **self-paced curriculum learning** pe dataset-ul "Vai de pielea mea", folosind CNN-ul ca model de baza. La fiecare epoca, se calculeaza loss-ul per exemplu pe intregul set de antrenare (fara gradient), iar modelul este antrenat doar pe exemplele cu loss sub un prag care creste progresiv.

Funcția de pacing: $q = \min(1.0, 0.3 + 0.07 * \text{epoch})$ — incepe cu 30% din date (cele mai usoare), ajunge la 100% dupa epoca 10. Exemplele sunt selectate dupa loss-ul curent al modelului, nu dupa o dificultate predefinita.

Comparatia se face cu un baseline identic (acelasi CNN, aceleasi date, fara filtrare).

8.2 Rezultate

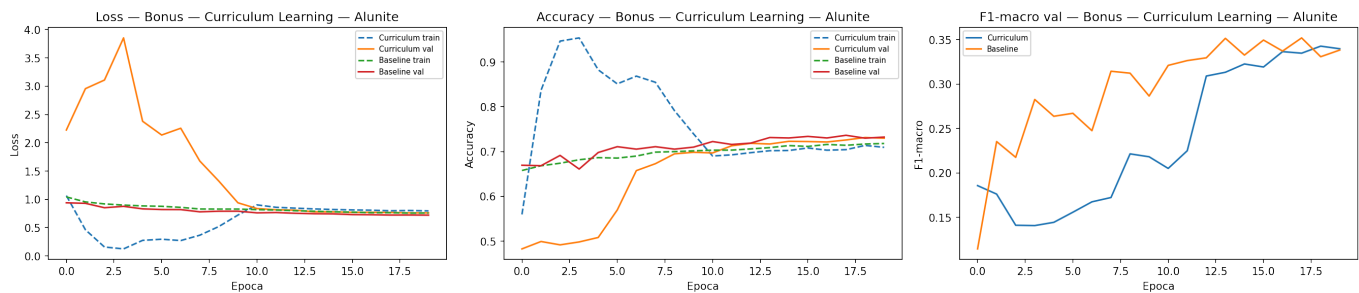


Fig. 25 — Curriculum Learning vs Baseline — CNN, Alunite

Experiment	Best val acc	Best F1-macro (ep 20)
CNN Curriculum	0.731	0.343
CNN Baseline	0.732	0.338

Observatie: Curriculum learning produce o convergenta diferita fata de baseline — in primele epoci F1-macro-ul este mai mic (modelul vede doar exemplele usoare), dar creste mai stabil pe masura ce pragul se ridica. La finalul antrenarii (ep 20), curriculum-ul obtine un F1-macro usor superior (0.343 vs 0.338), sugerand ca ordinea de prezentare a exemplurilor ajuta la generalizare, chiar daca diferenta este mica pe 20 de epoci. Un numar mai mare de epoci ar putea amplifica beneficiul.